

MNLP 2026 - Homework 1: Semantic Search and Fine-Tuning

Mohamed Aziz Selmi Samuele Costantinopoli

Sapienza University of Rome

{selmi.2264570, costantinopoli.2271799}@studenti.uniroma1.it

1 Introduction

This project addresses a semantic search task in a question-answering setting. Given a natural language query and a set of candidate text chunks, the goal is to retrieve the chunk containing the correct answer.

2 Methodology

To address the Retrieval via Semantic Search task, we explored different strategies, starting from simpler approaches and progressively moving toward more advanced ones.

2.1 Evaluation Metrics

Alongside Hit@K, we also used Mean Reciprocal Rank (MRR), which measures how highly the first correct answer is placed in the ranking:

$$MRR = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{rank_i}$$

2.2 Similarity Metrics

Besides cosine similarity, Euclidean and angular similarity were also evaluated to compare different ranking behaviours.

- **Euclidean Similarity**

$$similarity_{euc}(q, c) = 1 - \|q - c\|_2$$

- **Angular Distance**

$$similarity_{ang}(q, c) = 1 - \frac{\arccos(\cos_{sim}(q, c))}{\pi}$$

2.3 Fine-tuning Strategies

We tested several fine-tuning strategies to progressively improve retrieval quality.

- **DistilBERT with Triplet Loss:** Triplet loss was used as a simple contrastive baseline. Negative examples were sampled among the incorrect candidates of the same query. This choice was made after observing the dataset structure, where candidate answers within the same pool were often semantically close.

- **DistilBERT with InfoNCE:** We then adopted InfoNCE as a more advanced contrastive strategy. Negative examples were again sampled from the candidate pool of each query, allowing the model to compare the correct answer against multiple related candidates in the same step.

- **MiniLM (all-MiniLM-L6-v2) with Mixed Negatives:** We extended the InfoNCE strategy to MiniLM and introduced a mixed negative sampling strategy. Random negatives were combined with a smaller set of harder negatives selected through TF-IDF mining with `TfidfVectorizer`, applied over the same candidate pool. This choice was made to expose the model to both easy negatives, which improve general separation, and harder negatives, which force the encoder to learn finer semantic distinctions.

- **Fine-tuned BGE Re-ranker:** To enhance results on top of the fine-tuned models, `bge-reranker-base` was used as a second-stage re-ranker to refine retrieved candidates. It was fine-tuned as a cross-encoder on positive query-answer pairs and sampled negatives to further improve performance.

- **Fine-tuned BGE-large Model:** `bge-large-en-v1.5` was selected as a stronger reference model for comparison with the other approaches developed, and was fine-tuned with a simple triplet-loss strategy.

3 Experimental Setup

DistilBERT and MiniLM were implemented with HuggingFace Transformers and trained through custom PyTorch pipelines. Since both models were used to produce dense representations, the full ranking pipeline was built manually, including custom data structures for triplet and InfoNCE training (TripletSample and InfoNCEBatch), contrastive losses, and evaluation.

3.1 DistilBERT Triplet Fine-Tuning

DistilBERT (distilbert-base-uncased) was implemented with a custom triplet structure for the ranking task. The final setup used AdamW, cosine similarity, triplet margin 0.28, learning rate 2×10^{-5} , and three epochs.

3.2 DistilBERT InfoNCE Fine-Tuning

DistilBERT was implemented with a custom InfoNCE batch structure for contrastive ranking. Each query was paired with four negatives sampled from the candidate pool. The final setup used AdamW, Cross Entropy loss, temperature 0.07, learning rate 2×10^{-5} , angular similarity, and three epochs.

3.3 MiniLM InfoNCE Fine-Tuning

MiniLM (all-MiniLM-L6-v2) used a custom InfoNCE pipeline. Negative sampling combined eight random negatives with two hard negatives selected via TfidfVectorizer. Training used AdamW, batch size 32, Cross Entropy loss, temperature 0.04, learning rate 2×10^{-5} , cosine similarity, and two epochs.

3.4 Fine-Tuned BGE Re-ranker

The re-ranking stage was implemented with BAAI/bge-reranker-base using SentenceTransformers CrossEncoder. The model was fine-tuned on one positive pair and five sampled negatives per query, with batch size 16 and one epoch. Inference, the base model retrieves the top $k=12$ candidates, which the CrossEncoder then re-ranks.

3.5 Fine-Tuned BGE-large Model

BGE-large (BAAI/bge-large-en-v1.5) was implemented with SentenceTransformers and fine-tuned with triplet loss, cosine distance, triplet margin 0.30, learning rate 2×10^{-5} , batch size 32, and three epochs.

4 Results

The best-performing configurations are presented progressively.

- **DistilBERT Triplet Fine-Tuning:**

Fine-tuning DistilBERT with triplet loss produced a clear improvement over the untuned baseline, as shown in **Table 1**. Although simple, this setting proved effective, especially thanks to sampling negatives directly from the candidate pool. The final result also benefited from the triplet margin of 0.28, which empirically produced the most effective results.

- **DistilBERT InfoNCE Fine-Tuning:**

Fine-tuning DistilBERT with InfoNCE improved over the previous triplet-based setting, with final performance reported in **Table 2**. Comparing the correct answer against multiple candidate negatives at once led to stronger results.

- **MiniLM InfoNCE Fine-Tuning:**

MiniLM fine-tuned with InfoNCE produced the strongest result among the custom models, as shown in **Table 3**. This improvement was largely supported by the mixed negative strategy, where combining random negatives with a smaller set of harder TF-IDF negatives proved more effective than relying on a single type of negative.

- **MiniLM + BGE Re-ranker:**

MiniLM already achieved strong results, and adding the fine-tuned bge-reranker-base improved them further, as shown in **Table 4**. Since the initial ranking from the retrieval stage was already strong, fine-tuning the re-ranker became necessary to effectively refine the retrieved candidates without weakening the quality already achieved.

- **BGE-large Model:**

The BGE-large baseline already produced strong results, reported in **Table 5**, and fine-tuning with triplet loss improved them further, as shown in **Table 6**. Although MiniLM with mixed negatives alone did not surpass it, combining it with the fine-tuned BGE re-ranker led to an interesting result, allowing it to outperform the fine-tuned BGE-large model trained with triplet loss.

5 Appendices

Hit@1	0.5810
Hit@3	0.8040
Hit@5	0.8745
MRR	0.7094

Table 1: Performance - DistilBERT Triplet (Cosine)

Hit@1	0.6145
Hit@3	0.8485
Hit@5	0.9125
MRR	0.7430

Table 2: Performance - DistilBERT infoNCE (Angular)

Hit@1	0.6210
Hit@3	0.8565
Hit@5	0.9190
MRR	0.7510

Table 3: Performance - MiniLM InfoNCE (Cosine)

Hit@1	0.7040
Hit@3	0.8990
Hit@5	0.9505
MRR	0.8100

Table 4: Performance metrics for MiniLM InfoNCE reranker bge-reranker-base using Cosine similarity.

Hit@1	0.4955
Hit@3	0.8005
Hit@5	0.8845
MRR	0.6649

Table 5: Performance metrics for bge BaseLine using Cosine similarity.

Hit@1	0.6535
Hit@3	0.8535
Hit@5	0.9120
MRR	0.7664

Table 6: Performance metrics for bge Triplet using Cosine similarity.

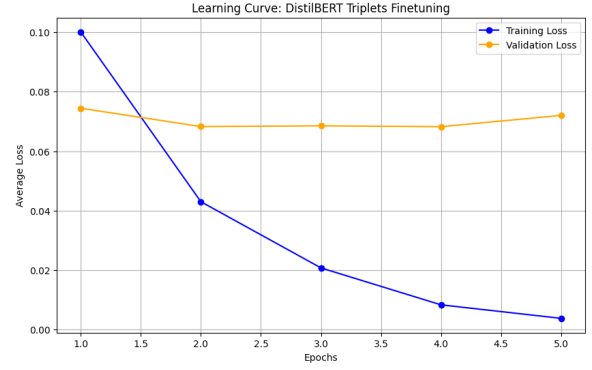


Figure 1: Training loss and validation loss DistilBERT triplet

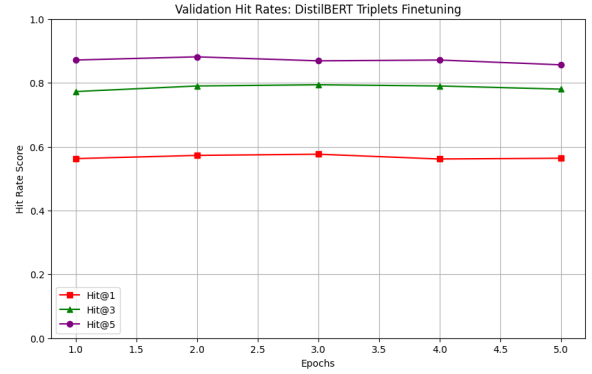


Figure 2: Hit Rate curves for DistilBERT triplet

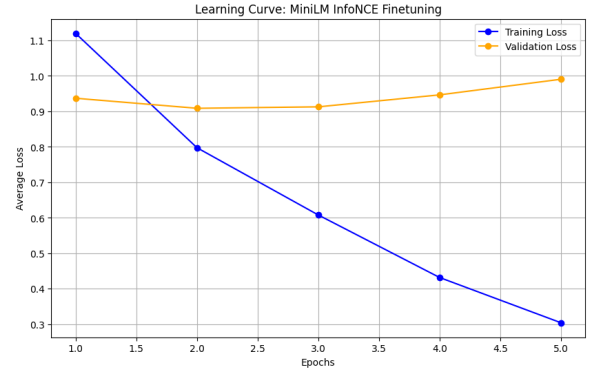


Figure 3: Training loss and validation loss MiniLM infoNCE

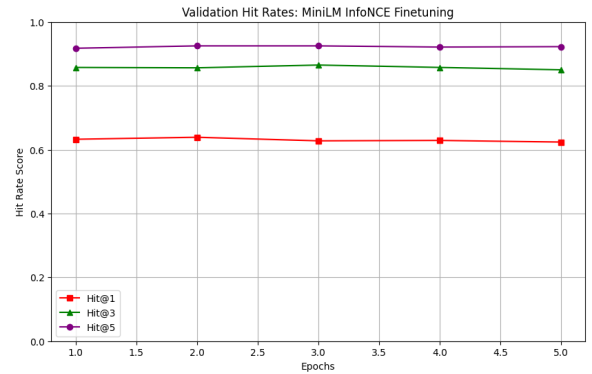


Figure 4: Hit Rate curves for MiniLM infoNCE